

CONFIGURE PORT SECURITY IN CISCO PACKET TRACER

AIM

To design a secure wired network topology in Cisco packet tracer using port security features (shutdown, restart, protect) and monitor unauthorized device access on switch interfaces.

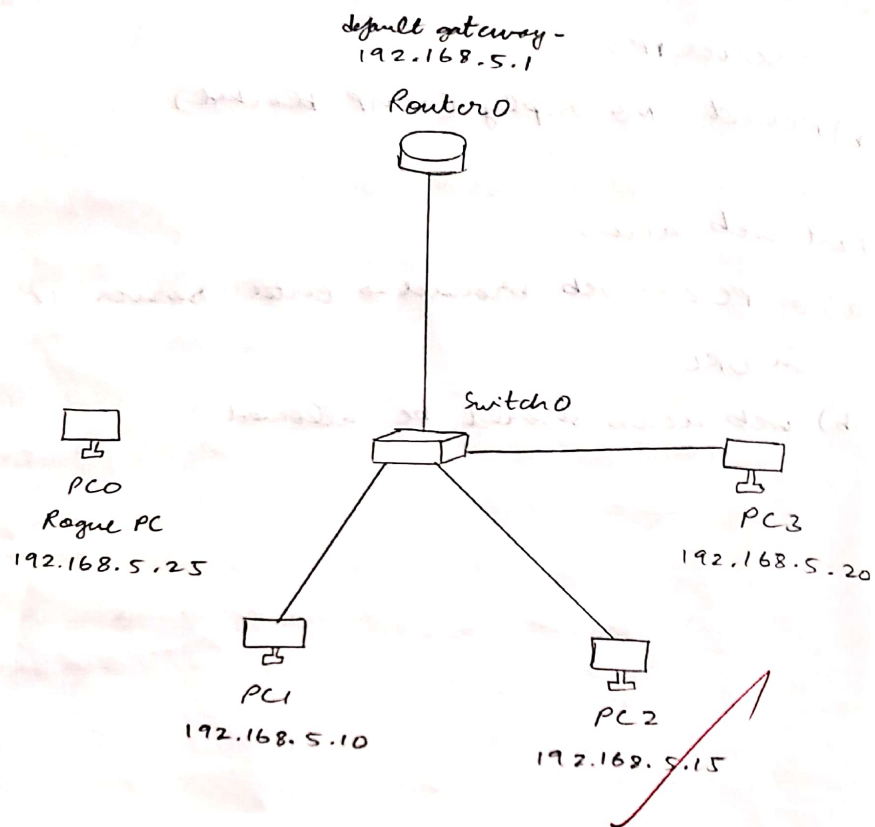
PROCEDURE

1. Create a single network topology with PCs and a router.
2. Connect PC1, PC2, PC3 to the router via a switch using wired cables.
3. Assign IP addresses and default gateways to all PC's and router.
4. Ping between PC's from command prompt to verify connectivity.

Port Security - shutdown mode:

5. Configure port security on switch ports fa0/1, fa1/1 and fa2/1 using shutdown violation mode.

6. Apply sticky MAC address, set max MAC to 1, and assign shutdown action on all switch ports.
7. Ping all PC's to confirm proper communication.
8. Assign IP to Rouge PC0.

TOPOLOGY

9. Disconnect PC1 and connect Rouge PC0, attempt to ping other PCs.
10. Ping PC2 from Rouge PC0 to trigger port shutdown.
11. Use "show" commands on switch to view port-security and MAC table.
12. Reconnect PC1 but observe red interference (Blocked)
13. Reactivate port using "shutdown" and "no shutdown" commands.
14. Verify connectivity from PC1 using ping.

Port-Security - Restrict-Mode:

- a) Configure port fa 1/1 to use "restrict" mode for violation.
- b) Replace PC2 with Rouge PC0.
- c) Attempt pings to verify time out, due to restriction.
- d) Check port-security status via "show" commands.
- e) ~~Reset~~ PC2 connection and test ping.

Port Security - Protect Mode:

- a) Configure port fa 2/1 to use "protect" mode for violations.
- b) Ping between PC's to confirm normal access.

OUTPUT

(i) Shutdown mode:

```
#ping 192.168.5.15
```

request timed out.

Request timed out.

Request timed out.

Request timed out.

Ping statistics for 192.168.5.15:

Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

(ii) Restrict mode:

```
C:\>ping 192.168.5.10
```

Pinging 192.168.5.10 with 32 bytes of data:

Request timed out.

Request timed out.

Request timed out.

Request timed out.

Ping statistics for 192.168.5.10:

Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

(iii) Protect mode:

C:\>ping 192.168.5.15

Pinging 192.168.5.15 with 32 bytes of data:

Request timed out.

Request timed out.

Request timed out.

Request timed out.

Ping statistics for 192.168.5.15

Packets: Sent=4, Received=0, Lost=4 (100% loss)



c) Replace PC3 with Rouge PC0.

d) attempt pings from Rouge PC0 to PCs.
(expect timeouts)

e) use show commands to monitor MAC
and port-security status.

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~~RESULT~~

Thus, using cisco packet tracer, port security
was successfully configured in shutdown,
restrict and protect modes to block
unauthorized access and secure the
network.